
Examiners' commentaries 2024 (October)

ST2134 Advanced statistics: statistical inference

Important note

This commentary reflects the examination and assessment arrangements for this course in the academic year 2023–24. The format and structure of the examination may change in future years, and any such changes will be publicised on the virtual learning environment (VLE).

Information about the subject guide and the Essential reading references

Unless otherwise stated, all cross-references will be to the latest version of the subject guide (2021). You should always attempt to use the most recent edition of any Essential reading textbook, even if the commentary and/or online reading list and/or subject guide refer to an earlier edition. If different editions of Essential reading are listed, please check the VLE for reading supplements – if none are available, please use the contents list and index of the new edition to find the relevant section.

General remarks

Learning outcomes

At the end of this half course and having completed the essential reading and activities you should be able to:

- explain the principles of data reduction
- judge the quality of estimators
- choose appropriate methods of statistical inference to tackle real problems.

Key steps to improvement

Candidates should pay close attention to the essential topics of sufficient statistics, point estimation, interval estimation and hypothesis testing. Being able to state and prove key theorems and lemmas simply requires close study of these in the subject guide. The point estimation techniques of method of moments estimation and maximum likelihood estimation are standard (and are even introduced in **ST104B Statistics 2** and **ST1215 Introduction to mathematical statistics**), with questions typically varying in terms of the probability distribution from which the random sample is drawn (with many common distributions covered in **ST104B Statistics 2**, **ST1215 Introduction to mathematical statistics** and **ST2133 Advanced statistics: distribution theory**). Pivotal functions feature heavily in Chapter 4 (Interval estimation), and likelihood ratio tests similarly feature heavily in Chapter 5 (Hypothesis testing).

An examination paper attempts to cover as much of the syllabus as possible, but inevitably some parts are left out. As the included parts will vary from year to year it is not enough to practise only

on the past papers. Candidates should practise on all the examples, activities and sample examination questions in the subject guide in order to understand the relevant theory and adequately prepare for the examination. Afterwards you may try one or two past papers for further practice.

The examination paper for this course involves four questions and each of the questions contains several parts which contribute towards the final grade with the different numbers of marks reflecting their difficulty. The difficulty of each part is usually assessed after taking into account the level of thinking related to statistical inference and the extent of algebraic computations required. Take 5–10 minutes to read the whole paper and try to identify the difficulties of each question in comparison to your ability. You will find that some questions could be much easier for you than others.

If you are unable to solve an exercise due to algebraic computations or lack of knowledge of the material of **ST2133 Advanced statistics: distribution theory**, do not abandon an exercise. Write clearly the procedure which should be followed for the relevant statistical inference task(s). Remember that the primary aspect you are being examined on is the ability to *understand* the statistical methodology.

A good knowledge of basic calculus, such as the ability to compute integrals, sums and derivatives, will give you invaluable help for completing the examination paper. Also, the main concepts of **ST2133 Advanced statistics: distribution theory** (random variables and their distribution, expectation, independence, conditional expectation etc.) should be well-understood before attempting this half course. If you are not comfortable with any of these, go back and revise. Memorising basic properties such as the moments and probability mass/density functions of some standard distributions (such as the binomial, Poisson, uniform, normal and gamma) is also a good idea.

When you are trying to prove an equality or inequality be careful of what is on the left-hand and right-hand sides. Justify the steps in the calculations as much as possible. If you cannot get a question right, do not just write in the correct answer without any further explanation. If you just write on the script what you think went wrong you will get more marks. Also, try to avoid writing more than you are asked to write. For example, when asked to *state* a theorem, do not give the proof as well as this simply wastes time.

Be clear in your answers. When giving the probability mass/density function of a random variable you should also state its sample space (also known as its support). When you are deriving an estimator or a sufficient statistic also state the parameter to which it corresponds. Sometimes it is a good idea to underline your final result (if applicable).

Make sure that you are able to provide the definitions of basic concepts such as estimators vs. estimates, confidence intervals and hypothesis tests. Remember that hypotheses are statements about population parameters which are estimated using functions of the data (that is, *statistics*). In the relevant calculations you should always be able to distinguish a statistic from a parameter and be clear on what they represent, respectively.

Practise operations involving products and sums. A substantial number of candidates fail to write them down correctly. Quite often they omit or write the indices incorrectly which inevitably leads to errors. A typical example of this is the following:

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population with probability density function $f_{X_i}(x_i; \theta)$, often written for simplicity as $f_X(x; \theta)$. An observed sample is $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. The fact that this sample is a random sample indicates that we can write the joint probability density function of $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta).$$

Make sure that you write the above and *not* the following:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_X(x; \theta)$$

which can lead to the *incorrect* expression:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = (f_X(x; \theta))^n.$$

Note that if you were asked to provide the log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$ the calculations would be formed of *sums* rather than *products* (refresh your memory on the properties of logarithmic functions). The correct expression would be:

$$\begin{aligned} l_{\mathbf{X}}(\theta; \mathbf{x}) &= \log(f_{\mathbf{X}}(\theta; \mathbf{x})) \\ &= \log \left(\prod_{i=1}^n f_{X_i}(x_i; \theta) \right) \\ &= \sum_{i=1}^n \log f_{X_i}(x_i; \theta). \end{aligned}$$

Examination revision strategy

Many candidates are disappointed to find that their examination performance is poorer than they expected. This may be due to a number of reasons, but one particular failing is ‘**question spotting**’, that is, confining your examination preparation to a few questions and/or topics which have come up in past papers for the course. This can have serious consequences.

We recognise that candidates might not cover all topics in the syllabus in the same depth, but you need to be aware that examiners are free to set questions on **any aspect** of the syllabus. This means that you need to study enough of the syllabus to enable you to answer the required number of examination questions.

The syllabus can be found in the Course information sheet available on the VLE. You should read the syllabus carefully and ensure that you cover sufficient material in preparation for the examination. Examiners will vary the topics and questions from year to year and may well set questions that have not appeared in past papers. Examination papers may legitimately include questions on any topic in the syllabus. So, although past papers can be helpful during your revision, you cannot assume that topics or specific questions that have come up in past examinations will occur again.

If you rely on a question-spotting strategy, it is likely you will find yourself in difficulties when you sit the examination. We strongly advise you not to adopt this strategy.

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Comments on specific questions

Candidates should answer all **FOUR** questions: **Question 1** of Section A (40 marks) and all **THREE** questions from Section B (60 marks in total). **Candidates are strongly advised to divide their time accordingly.**

Section A

Answer Question 1 from this section.

Question 1

- (a) State and prove the Cramér–Rao inequality theorem.

(10 marks)

Reading for this question

Section 3.8 of the subject guide discusses the variance of unbiased estimators Cramér–Rao lower bound.

Approaching the question

We consider the continuous case. We make use of the variance/covariance form of the Cauchy–Schwarz inequality which states that for two random variables X and Y we have:

$$(\text{Cov}(X, Y))^2 \leq \text{Var}(X) \text{Var}(Y).$$

By unbiasedness of $T(\mathbf{X})$, we have:

$$E(T(\mathbf{X})) = g(\theta)$$

hence:

$$g(\theta) = \int_{\mathbb{R}^n} T(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x}.$$

Differentiating both sides with respect to θ gives us:

$$\frac{d}{d\theta} g(\theta) = \int_{\mathbb{R}^n} T(\mathbf{x}) \frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} = \int_{\mathbb{R}^n} T(\mathbf{x}) s_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} = E(T(\mathbf{X}) s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Since $E(s_{\mathbf{X}}(\theta; \mathbf{X})) = 0$, we have:

$$\begin{aligned} \text{Cov}(T(\mathbf{X}), s_{\mathbf{X}}(\theta; \mathbf{X})) &= E(T(\mathbf{X}) s_{\mathbf{X}}(\theta; \mathbf{X})) - E(T(\mathbf{X})) E(s_{\mathbf{X}}(\theta; \mathbf{X})) \\ &= E(T(\mathbf{X}) s_{\mathbf{X}}(\theta; \mathbf{X})) \end{aligned}$$

hence we may write:

$$\frac{d}{d\theta} g(\theta) = \text{Cov}(T(\mathbf{X}), s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Applying the Cauchy–Schwarz inequality then yields:

$$\left(\frac{d}{d\theta} g(\theta) \right)^2 \leq \text{Var}(T(\mathbf{X})) \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Rearranging, we have:

$$\text{Var}(T(\mathbf{X})) \geq \frac{\left(\frac{d}{d\theta} g(\theta) \right)^2}{\text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X}))} = \frac{\left(\frac{d}{d\theta} g(\theta) \right)^2}{\mathcal{I}_{\mathbf{X}}(\theta)}$$

since $\mathcal{I}_{\mathbf{X}}(\theta) = \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X}))$.

- (b) Let X_1 and X_2 be two independent random variables with the same mean, μ , and the same finite variance, $\sigma^2 < \infty$. Let $\hat{\mu} = \alpha X_1 + \beta X_2$ be an estimator of μ , where α and β are two non-zero constants.

- i. Identify the condition on α and β to ensure that $\hat{\mu}$ is an unbiased estimator of μ .

(3 marks)

- ii. Find the estimator with the minimum mean squared error among *all* unbiased linear estimators $\hat{\mu}$ of μ . You do *not* need to show that the stationary point is a minimum.

(5 marks)

Reading for this question

Section 3.4 of the subject guide covers relevant material on the statistical properties of point estimators.

Approaching the question

- i. Let:

$$E(\hat{\mu}) = E(\alpha X_1 + \beta X_2) = \alpha E(X_1) + \beta E(X_2) = (\alpha + \beta)\mu.$$

Hence $\alpha + \beta = 1$ is the condition for $\hat{\mu}$ to be an unbiased estimator of μ .

- ii. Under this condition, noting that $\beta = 1 - \alpha$, we have:

$$\begin{aligned} \text{MSE}(\hat{\mu}) &= \text{Var}(\hat{\mu}) = \alpha^2 \text{Var}(X_1) + \beta^2 \text{Var}(X_2) = (\alpha^2 + \beta^2)\sigma^2 \\ &= (2\alpha^2 - 2\alpha + 1)\sigma^2. \end{aligned}$$

Setting:

$$\frac{d}{d\alpha} \text{MSE}(\hat{\mu}) = (4\alpha - 2)\sigma^2 = 0$$

we have $\alpha = 0.50$, and hence $\beta = 0.50$. Therefore, among all unbiased linear estimators, the sample mean $\bar{X} = (X_1 + X_2)/2$ has the minimum variance.

- (c) In a bag, there are only blue and green marbles. A random sample of size 5, drawn with replacement, results in the sequence:

blue, green, blue, green, green.

The precise count of blue and green marbles in the bag is uncertain, but the proportion, π , of blue marbles is either 0.25 or 0.50. What is the maximum likelihood estimate of π ?

(6 marks)

Reading for this question

Section 3.9 of the subject guide covers maximum likelihood estimation.

Approaching the question

The joint probability, and hence likelihood function, of the observed sequence of marbles is:

$$L_{\mathbf{X}}(\pi; \mathbf{x}) = \pi(1 - \pi)\pi(1 - \pi)(1 - \pi) = \pi^2(1 - \pi)^3.$$

For $\pi = 0.25$, we have:

$$L_{\mathbf{X}}(0.25; \mathbf{x}) = (0.25)^2(0.75)^3 = 0.0264$$

which is less than when $\pi = 0.50$, since:

$$L_{\mathbf{X}}(0.50; \mathbf{x}) = (0.50)^5 = 0.03125.$$

Therefore, $\hat{\pi} = 0.50$.

Note that since there are only two possible values of π , there is no need to derive the log-likelihood function as it is straightforward to enumerate $L_{\mathbf{X}}(0.25; \mathbf{x})$ and $L_{\mathbf{X}}(0.50; \mathbf{x})$.

- (d) For an estimator $\hat{\theta}$ of θ , show that its mean squared error, $E((\hat{\theta} - \theta)^2)$, can be expressed as:

$$\text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2.$$

(6 marks)

Reading for this question

Section 3.4.1 of the subject guide presents the decomposition of the mean squared error.

Approaching the question

We have:

$$\begin{aligned} \text{MSE}_{\theta}(\hat{\theta}) &= E((\hat{\theta} - \theta)^2) \\ &= E(\underbrace{(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)}_{=0})^2 \\ &= E((\hat{\theta} - E(\hat{\theta}))^2 + (E(\hat{\theta}) - \theta)^2 + 2(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta)) \\ &= E((\hat{\theta} - E(\hat{\theta}))^2) + E((\text{Bias}(\hat{\theta}))^2) + 2 \underbrace{(E(\hat{\theta}) - E(\hat{\theta}))(E(\hat{\theta}) - \theta)}_{=0} \\ &= \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2. \end{aligned}$$

- (e) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the $N(0, \sigma^2)$ distribution. Find a pivotal function for the construction of a confidence interval for σ^2 . Use the pivotal function to construct a confidence interval with a confidence level of $100(1 - \alpha)\%$.

(10 marks)

Reading for this question

Section 4.6 of the subject guide explains how to find interval estimators from pivotal functions.

Approaching the question

To find a confidence interval for σ^2 when sampling from the normal distribution we proceed by standardising and squaring to obtain a χ^2 distribution. Since each:

$$\frac{X_i}{\sigma} \sim N(0, 1)$$

and the X_i s are independent, by squaring and summing we get:

$$\frac{\sum_{i=1}^n X_i^2}{\sigma^2} \sim \chi_n^2.$$

The function $\sum_{i=1}^n X_i^2 / \sigma^2$ is a pivotal function for σ^2 .

As $\sum_{i=1}^n X_i^2 / \sigma^2 \sim \chi_n^2$, we can write:

$$1 - \alpha = P \left(\chi_{n, 1-\alpha/2}^2 < \frac{\sum_{i=1}^n X_i^2}{\sigma^2} < \chi_{n, \alpha/2}^2 \right) = P \left(\frac{\sum_{i=1}^n X_i^2}{\chi_{n, \alpha/2}^2} < \sigma^2 < \frac{\sum_{i=1}^n X_i^2}{\chi_{n, 1-\alpha/2}^2} \right)$$

suggesting that an exact $100(1 - \alpha)\%$ confidence level interval for σ^2 is the following:

$$\left[\frac{\sum_{i=1}^n X_i^2}{\chi_{n, \alpha/2}^2}, \frac{\sum_{i=1}^n X_i^2}{\chi_{n, 1-\alpha/2}^2} \right].$$

Section B

Answer all three questions from this section.

Question 2

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a distribution with the probability density function:

$$f_X(x; \theta) = \begin{cases} 2\theta^{-2}x & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is an unknown parameter.

- (a) Let $Y = X_{(n)}$, i.e. the sample maximum. Determine the probability density function of Y .

(4 marks)

- (b) Find a simple sufficient statistic for θ .

(4 marks)

- (c) Find the maximum likelihood estimator, $\hat{\theta}$, of θ .

(6 marks)

(d) Compute the mean squared error of $\hat{\theta}$.

(6 marks)

Reading for this question

Sections 2.8 and 3.9 of the subject guide cover relevant material on sufficient statistics and maximum likelihood estimators, respectively.

Approaching the question

(a) For $x \in [0, \theta]$, we have:

$$F_X(x; \theta) = \int_0^x 2\theta^{-2}t \, dt = \left[\frac{t^2}{\theta^2} \right]_0^x = \frac{x^2}{\theta^2}.$$

Hence the probability density function of Y is:

$$f_Y(y; \theta) = n(F_X(y; \theta))^{n-1} f_X(y; \theta) = n \frac{y^{2n-2}}{\theta^{2n-2}} \frac{2y}{\theta^2} = \frac{2ny^{2n-1}}{\theta^{2n}}$$

for $0 \leq y \leq \theta$, and 0 otherwise.

(b) The joint pdf is:

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}; \theta) &= \frac{2^n}{\theta^{2n}} \prod_{i=1}^n x_i I(0 \leq x_{(1)} \leq x_{(n)} \leq \theta) \\ &= \frac{1}{\theta^{2n}} I(x_{(n)} \leq \theta) 2^n \prod_{i=1}^n x_i I(x_{(1)} \geq 0). \end{aligned}$$

Hence, by the factorisation theorem, we have:

$$g(x_{(n)}, \theta) = \frac{1}{\theta^{2n}} I(x_{(n)} \leq \theta) \quad \text{and} \quad h(\mathbf{x}) = 2^n \prod_{i=1}^n x_i I(x_{(1)} \geq 0).$$

and so $X_{(n)}$ is a sufficient statistic for θ .

(c) From part (b), the likelihood function $L_{\mathbf{X}}(\theta; \mathbf{x})$ is 0 when $\theta < X_{(n)}$. Also, we note that for $\theta \geq X_{(n)}$, $L_{\mathbf{X}}(\theta; \mathbf{x})$ decreases as θ increases. Hence $L_{\mathbf{X}}(\theta; \mathbf{x})$ obtains its maximum at $X_{(n)}$, i.e. $\hat{\theta} = X_{(n)}$ is the maximum likelihood estimator of θ .

(d) We can compute the following quantities. First:

$$E(\hat{\theta}) = \int_0^\theta 2n \frac{y^{2n}}{\theta^{2n}} \, dy = \frac{2n\theta}{2n+1}$$

hence the bias of $\hat{\theta}$ is:

$$E(\hat{\theta}) - \theta = \frac{2n\theta}{2n+1} - \theta = -\frac{\theta}{2n+1}.$$

Also:

$$E(\hat{\theta}^2) = \int_0^\theta 2n \frac{y^{2n+1}}{\theta^{2n}} \, dy = \frac{2n\theta^2}{2n+2}$$

hence:

$$\text{Var}(\hat{\theta}) = \frac{2n}{2n+2} \theta^2 - \left(\frac{2n}{2n+1} \right)^2 \theta^2 = \frac{2n\theta^2}{(2n+2)(2n+1)^2}.$$

Therefore:

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2 \\ &= \frac{2n\theta^2}{(2n+2)(2n+1)^2} + \left(-\frac{\theta}{2n+1} \right)^2 \\ &= \frac{4n+2}{(2n+2)(2n+1)^2} \theta^2. \end{aligned}$$

Question 3

- (a) Let $\mathbf{X} = \{X_1, X_2, \dots, X_{10}\}$ be a random sample from a distribution with mean μ . Suppose $\sum_{i=1}^{10} X_i$ is a sufficient statistic for μ . Based on the initial estimator $\hat{\mu} = (X_1 + X_2)/2$, use the Rao–Blackwell theorem to find an improved estimator of μ .

(6 marks)

Reading for this question

Section 3.5 of the subject guide covers the Rao–Blackwell theorem in terms of the application of sufficiency in point estimation.

Approaching the question

Since $X_1, X_2, \dots, X_{10}$ are identically distributed:

$$E\left(X_i \mid \sum_{i=1}^{10} X_i\right) = E\left(X_1 \mid \sum_{i=1}^{10} X_i\right).$$

Using the Rao–Blackwell theorem, if we condition on the sufficient statistic $\sum_{i=1}^{10} X_i$, we get the following improved estimator:

$$\begin{aligned} E\left(\hat{\mu} \mid \sum_{i=1}^{10} X_i\right) &= E\left(\frac{X_1 + X_2}{2} \mid \sum_{i=1}^{10} X_i\right) = \frac{1}{2} E\left(X_1 \mid \sum_{i=1}^{10} X_i\right) + \frac{1}{2} E\left(X_2 \mid \sum_{i=1}^{10} X_i\right) \\ &= E\left(X_1 \mid \sum_{i=1}^{10} X_i\right) \\ &= E\left(X_i \mid \sum_{i=1}^{10} X_i\right) \\ &= \frac{1}{10} \sum_{i=1}^{10} E\left(X_i \mid \sum_{i=1}^{10} X_i\right) \\ &= \frac{1}{10} E\left(\sum_{i=1}^{10} X_i \mid \sum_{i=1}^{10} X_i\right) \\ &= \frac{1}{10} \sum_{i=1}^{10} X_i \\ &= \bar{X}. \end{aligned}$$

So the improved estimator of μ is $\bar{X}$.

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the distribution with the probability density function:

$$f_X(x; \mu) = \left(\frac{1}{2\pi x^3}\right)^{1/2} \exp\left(\frac{1}{\mu} - \frac{x}{2\mu^2} - \frac{1}{2x}\right) \quad \text{for } x > 0$$

and 0 otherwise, where $\mu > 0$ is an unknown parameter.

- i. Derive a one-dimensional sufficient statistic for μ .

(4 marks)

- ii. Determine the maximum likelihood estimator of μ .

(4 marks)

- iii. Find an approximate pivotal function for μ and construct an asymptotic $100(1 - \alpha)\%$ confidence interval for μ with $\alpha \in (0, 1)$.

(6 marks)

Reading for this question

Sections 2.8, 3.9 and 4.6 of the subject guide cover relevant material on sufficient statistics, maximum likelihood estimators, and pivotal functions, respectively.

Approaching the question

- i. The likelihood function is:

$$\begin{aligned} L_{\mathbf{X}}(\mu; \mathbf{x}) &= \prod_{i=1}^n f_X(x_i; \mu) = \prod_{i=1}^n \left(\frac{1}{2\pi x_i^3} \right)^{1/2} \exp \left(\frac{1}{\mu} - \frac{x_i}{2\mu^2} - \frac{1}{2x_i} \right) \\ &= \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2} \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} - \frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \\ &= \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right) \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2}. \end{aligned}$$

We can write $f_{\mathbf{X}}(\mathbf{x}; \mu) = g \left(\sum_{i=1}^n x_i, \mu \right) h(\mathbf{x})$ by setting:

$$g \left(\sum_{i=1}^n x_i, \mu \right) = \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right)$$

and:

$$h(\mathbf{x}) = \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2}.$$

Hence, by the factorisation theorem, $\sum_{i=1}^n X_i$ is a one-dimensional sufficient statistic for μ .

- ii. Note that the likelihood function is proportional to:

$$L_{\mathbf{X}}(\mu; \mathbf{x}) \propto \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right).$$

The log-likelihood function can therefore be written as:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = \log L_{\mathbf{X}}(\mu; \mathbf{x}) \propto \frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2}.$$

The score function is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{\partial}{\partial \mu} l_{\mathbf{X}}(\mu; \mathbf{x}) = -\frac{n}{\mu^2} + \frac{\sum_{i=1}^n x_i}{\mu^3}.$$

Setting $s_{\mathbf{X}}(\mu; \mathbf{x}) = 0$ provides $\hat{\mu} = \sum_{i=1}^n X_i / n = \bar{X}$ as a candidate maximum likelihood estimator. Since:

$$\left. \frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) \right|_{\mu=\bar{x}} = \frac{2n}{\mu^3} - \frac{3n\bar{x}}{\mu^4} \Big|_{\mu=\bar{x}} = \frac{2n}{\bar{x}^3} - \frac{3n\bar{x}}{\bar{x}^4} = -\frac{n}{\bar{x}^3} < 0$$

we conclude that this is indeed the maximum likelihood estimator.

iii. We know that for large sample sizes:

$$\hat{\mu}_{ML} \sim N\left(\mu, \frac{1}{\mathcal{I}_{\mathbf{X}}(\mu)}\right)$$

approximately. In an extra level of approximation, we may use:

$$\hat{\mu}_{ML} \sim N\left(\mu, \frac{1}{H(\mu; \mathbf{x})}\right)$$

approximately, where:

$$H(\mu; \mathbf{x}) = - \frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) \Big|_{\mu=\bar{x}} = \frac{n}{\bar{x}^3}.$$

Let $S_{\mu} = \bar{x}^3/n$. We can then write:

$$\frac{\hat{\mu} - \mu}{\sqrt{S_{\mu}}} \sim N(0, 1)$$

approximately. Using the pivotal function technique for constructing confidence intervals, we get the following asymptotic $100(1 - \alpha)\%$ confidence interval:

$$\left(\hat{\mu} - z_{\alpha/2}\sqrt{S_{\mu}}, \hat{\mu} + z_{\alpha/2}\sqrt{S_{\mu}}\right)$$

where $z_{\alpha/2}$ is the top 100α th percentile of $N(0, 1)$.

Question 4

(a) State the Neyman–Pearson lemma.

(6 marks)

Reading for this question

Section 5.5.1 of the subject guide presents the Neyman–Pearson lemma.

Approaching the question

Consider a test procedure T of size α for $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $H_1 : \boldsymbol{\theta} = \boldsymbol{\theta}_1$. If the rejection region for T is:

$$R_T = \{\mathbf{x} \in \mathbb{R}^n : L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_{\alpha} L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x}) > 0\}$$

equivalently:

$$R_T = \left\{ \mathbf{x} \in \mathbb{R}^n : \frac{L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x})}{L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})} > k_{\alpha} \right\}$$

then T is the most powerful test of size α .

Here $L_{\mathbf{X}}$ is the likelihood function and k_{α} is a constant dependent on α . Note that $\alpha = P_{\boldsymbol{\theta}_0}(\mathbf{X} \in R_T)$.

(b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ and $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_n\}$ be two independent random samples from two exponential distributions with means μ_1 and μ_2 , respectively.

i. Find the maximum likelihood estimators of μ_1 , μ_2 , and also of μ when $\mu_1 = \mu_2 = \mu$. You do *not* need to show the stationary points are maxima.

(6 marks)

ii. Provide the likelihood ratio test of:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_1 : \mu_1 \neq \mu_2.$$

(5 marks)

- iii. Specify the asymptotic distribution under H_0 of the test statistic that you determined in part ii., and state the condition for rejecting H_0 .

(3 marks)

Reading for this question

Sections 5.6 and 5.8 of the subject guide present relevant material on likelihood ratio tests and properties of the likelihood ratio, respectively.

Approaching the question

- i. The joint probability density function of the two independent random samples is:

$$f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}; \mu_1, \mu_2) = \mu_1^{-n} \exp\left(-\sum_{i=1}^n \frac{x_i}{\mu_1}\right) \mu_2^{-n} \exp\left(-\sum_{i=1}^n \frac{y_i}{\mu_2}\right).$$

The (unconstrained) maximum likelihood estimators are:

$$\hat{\mu}_1 = \bar{X} \quad \text{and} \quad \hat{\mu}_2 = \bar{Y}.$$

Under $H_0 : \mu_1 = \mu_2 = \mu$, the (constrained) maximum likelihood estimator is:

$$\hat{\mu} = \frac{\bar{X} + \bar{Y}}{2}.$$

- ii. The likelihood ratio statistic is:

$$LR(\mathbf{X}, \mathbf{Y}) = \frac{L(\hat{\mu}_1, \hat{\mu}_2; \mathbf{x}, \mathbf{y})}{L(\hat{\mu}; \mathbf{x}, \mathbf{y})} = \frac{(\bar{X} + \bar{Y})^{2n} / 2^{2n}}{\bar{X}^n \bar{Y}^n} = 2^{-2n} \left(\sqrt{\frac{\bar{X}}{\bar{Y}}} + \sqrt{\frac{\bar{Y}}{\bar{X}}} \right)^{2n}.$$

- iii. Under H_0 , we have:

$$2 \log LR(\mathbf{X}, \mathbf{Y}) = 4n \log \left(\sqrt{\frac{\bar{X}}{\bar{Y}}} + \sqrt{\frac{\bar{Y}}{\bar{X}}} \right) - 4n \log 2 \sim \chi_1^2$$

approximately. We reject H_0 if $2 \log LR(\mathbf{x}, \mathbf{y}) > \chi_{\alpha, 1}^2$, where $\chi_{\alpha, 1}^2$ denotes the top 100α th percentile of the χ_1^2 distribution.